GLOBAL SOLAR ATLAS
BY WORLD BANK GROUP

Phoenix

33.543171°, -112.073593°
Murphy Bridle Path, Phoenix, Arizona, United States of America
Time zone: UTC-07, America/Phoenix [MST]

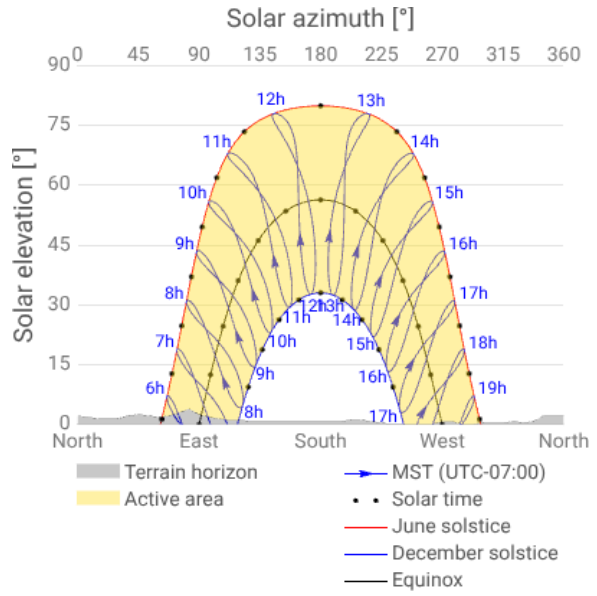
Report generated: 18 Jul 2026

SITE INFO

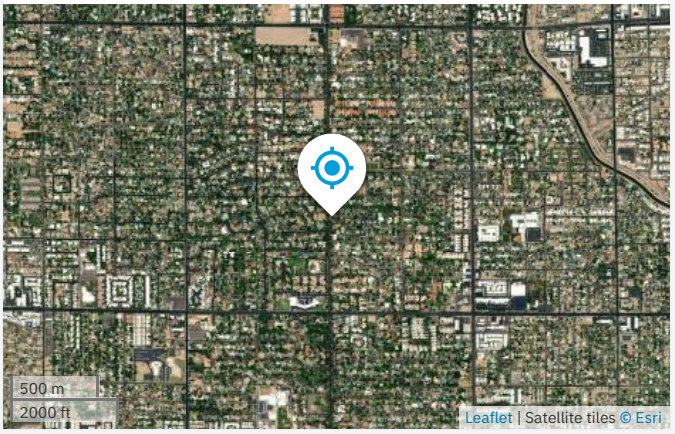
Map data

		Per year	
Specific photovoltaic power output	PVOUT specific	**1893.4**	kWh/kWp
Direct normal irradiation	DNI	**2719.8**	kWh/m^2
Global horizontal irradiation	GHI	**2119.4**	kWh/m^2
Diffuse horizontal irradiation	DIF	**478.4**	kWh/m^2
Global tilted irradiation at optimum angle	GTI opta	**2440.5**	kWh/m^2
Optimum tilt of PV modules	OPTA	**32 / 180**	°
Air temperature	TEMP	**23.1**	°C
Terrain elevation	ELE	**369**	m

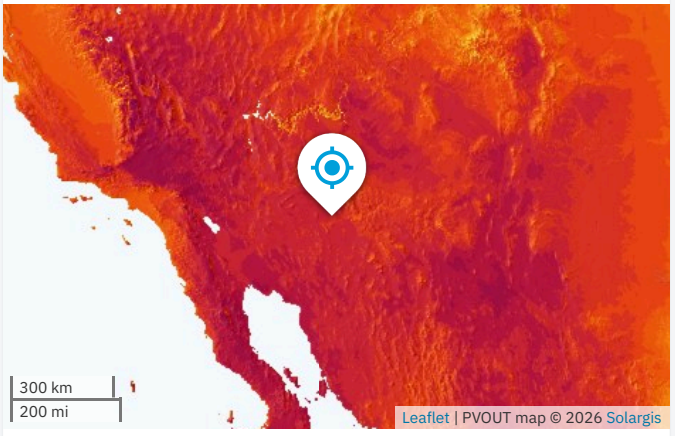
Horizon and sunpath



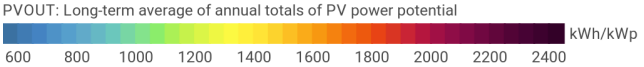
Map



PVOUT map



PVOUT: Long-term average of annual totals of PV power potential



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PV ELECTRICITY AND SOLAR RADIATION

Annual averages

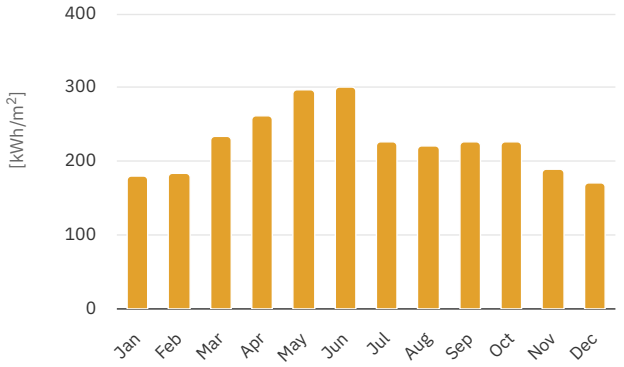
Direct normal irradiation

2710.2

kWh/m^2 per year

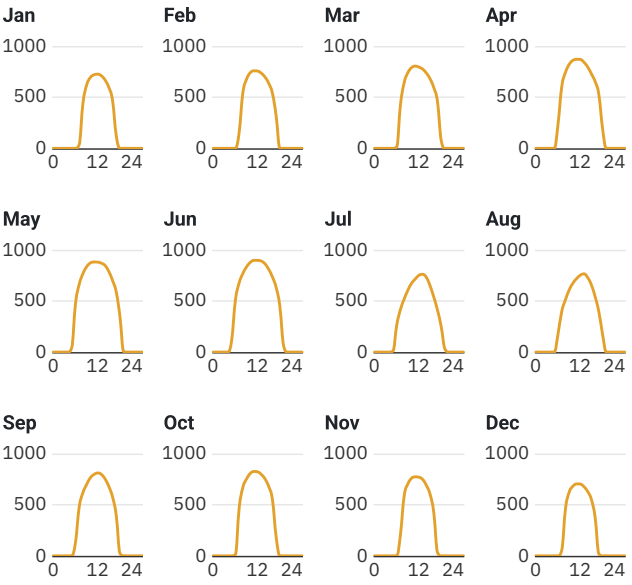
Monthly averages

Direct normal irradiation



Average hourly profiles

Direct normal irradiation [Wh/m^2]



UTC-07

Average hourly profiles

Direct normal irradiation [Wh/m^2]

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
0 - 1												
1 - 2												
2 - 3												
3 - 4												
4 - 5												
5 - 6					44	117	10					
6 - 7			14	240	471	506	264	179	130	12		
7 - 8	35	151	424	603	662	675	416	400	475	461	207	56
8 - 9	436	539	641	743	766	767	518	512	616	668	584	462
9 - 10	617	674	739	821	834	832	604	607	696	751	703	623
10 - 11	692	739	789	860	871	877	673	673	763	805	758	681
11 - 12	719	758	804	870	881	897	714	718	796	825	773	704
12 - 13	725	755	793	869	879	897	751	751	810	821	769	702
13 - 14	709	734	774	841	869	889	760	765	796	796	747	675
14 - 15	667	696	735	795	846	862	730	728	751	752	689	628
15 - 16	595	635	679	727	788	808	648	656	678	676	601	549
16 - 17	469	537	593	653	700	734	537	541	576	548	421	375
17 - 18	124	315	454	527	587	631	410	395	396	182	33	20
18 - 19			76	193	346	445	248	174	25			
19 - 20					4	63	30					
20 - 21												
21 - 22												
22 - 23												
23 - 24												
Sum	5,787	6,532	7,516	8,742	9,548	10,001	7,313	7,100	7,509	7,298	6,285	5,475

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GLOSSARY

Acronym	Full name	Unit	Type of use
DIF	Diffuse horizontal irradiation	kWh/m², MJ/m²	Average yearly, monthly or daily sum of diffuse horizontal irradiation (© 2025 Solargis)
DNI	Direct normal irradiation	kWh/m², MJ/m²	Average yearly, monthly or daily sum of direct normal irradiation (© 2025 Solargis)
ELE	Terrain elevation	m, ft	Elevation of terrain surface above/below sea level, processed and integrated from SRTM-3 data and related data products (SRTM v4.1 © 2004 - 2025, CGIAR-CSI)
GHI	Global horizontal irradiation	kWh/m², MJ/m²	Average annual, monthly or daily sum of global horizontal irradiation (© 2025 Solargis)
GTI	Global tilted irradiation	kWh/m², MJ/m²	Average annual, monthly or daily sum of global tilted irradiation (© 2025 Solargis)
GTI_opta	Global tilted irradiation at optimum angle	kWh/m², MJ/m²	Average annual, monthly or daily sum of global tilted irradiation for PV modules fix-mounted at optimum angle (© 2025 Solargis)
OPTA	Optimum tilt of PV modules	°	Optimum tilt of fix-mounted PV modules facing towards Equator set for maximizing GTI input (© 2025 Solargis)
PVOUT_total	Total photovoltaic power output	kWh, MWh, GWh	Yearly and monthly average values of photovoltaic electricity (AC) delivered by the total installed capacity of a PV system (© 2025 Solargis)
PVOUT_specific	Specific photovoltaic power output	kWh/kWp	Yearly and monthly average values of photovoltaic electricity (AC) delivered by a PV system and normalized to 1 kWp of installed capacity (© 2025 Solargis)
TEMP	Air temperature	°C, °F	Average yearly, monthly and daily air temperature at 2 m above ground. Calculated from outputs of ERA5 model (© 2025 ECMWF, post-processed by Solargis)

ABOUT

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